

Sparse trip demand prediction for shared E-scooter using spatio-temporal graph neural networks

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ABSTRACT

The shared electric scooter (E-scooter) is an emerging micro-mobility mode in sustainable cities. Accurate hourly trip demand prediction is critical for effective service maintenance, but it poses a challenge due to the dynamic distribution influenced by urban complexity. We propose a model, the Sparse Diffusion Convolutional Gated Recurrent Unit (SpDCGRU), which incorporates diffusion convolution layers into the gated recurrent unit (GRU) model, enabling the simultaneous capture of spatio-temporal dependencies. Tackling the data in Louisville, Kentucky, USA, we demonstrate that spatial data reclustering and fusion loss training strategies contribute to the prediction performance. Moreover, the periodic and weather features positively impact predicting the low and high trip demand levels, respectively. Our model outperforms others in terms of overall performance and each trip demand level, with a 4.75% improvement in the mean absolute error (MAE) compared to the graph convolutional recurrent network (GCRN).

1. Introduction

As urban sustainability and transit-oriented development have become crucial issues, the emerging shared E-scooter service is considered one of the potential strategies for addressing these. It is a novel individual short-term travel service based on mobile phones through the integration of a global positioning system (GPS), Internet of Things (IoT), and application programming interface (API) technologies (Wanganoo et al., 2022). This on-demand and dockless service is convenient, flexible, and time-saving and causes reduced carbon emissions using electricity-driven motors (Shaheen et al., 2020). Some researchers state that it also discourages citizens' reliance on private vehicles for short-term trips (Abduljabbar et al., 2021) and increases public transportation usage by providing first-and-last-mile connectivity (Abduljabbar et al., 2021; Witzel, 2018). From 2017 to 2022, it has already become more widespread in the U.S. and had tremendous growth compared to shared dock bike services. According to the (Bureau of Transportation Statistics, Department of Transportation, U.S., 2022), the number of shared E-scooter systems has grown to 300 in 158 cities, approximately double that of shared bike systems.

Related works have been conducted, with one of the fundamental tasks being demand prediction. In the overall urban scheme, city planners can take advantage of it to improve urban planning and public transportation accordingly. In the on-demand scheme, shared E-scooter operators require real-time demand forecasting to conduct more promising supply and maintenance management, known as reallocation strategies, to balance supply and demand. It can reduce redundant fleets with unnecessary maintenance costs in off-peak hours while enhancing the service capability to cover more customer usage in peak hours and popular regions. Furthermore, it also contributes to optimizing planning battery-swapping systems (Altintasi and Yalcinkaya, 2022; Chen et al., 2018). However, the prediction of shared E-scooter demand is sophisticated and challenging because of:

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- *Spatio-temporal trip demands with extreme sparsity and imbalance.* The shared E-scooter, an emerging mode of mobility, appeals to specific user groups (Jiao and Bai, 2020; Huo et al., 2021; Caspi et al., 2020), like college students, for short-term usage. Its ridership is influenced by weather conditions (Mathew et al.; Zou et al., 2020; Zhu et al., 2020; Noland, 2019) and periodic characteristics (Zhu et al., 2020; Heumann et al., 2021; McKenzie, 2020). As a result, its trip demand pattern is likely sparse and imbalanced over periods and regions in some cities (Chu et al., 2022). When most regions and times show minimal or near-zero demand values, accurately forecasting the few but significant high-demand values is crucial. In practice, shared E-scooter service operators prioritize identifying regions with higher demand during specific periods. This allows them to efficiently relocate underused scooters from regions of low demand to meet needs strategically.
- *Difficulties in predicting complexity and heterogeneity.* Simply relying on trip historical records is insufficient to predict travel behaviors accurately. Other integrated spatio-temporal data such as built environments (Bai and Jiao, 2020; Huo et al., 2021), weather conditions (Mathew et al.; Zhu et al., 2020; Noland, 2019), and periodic features (Zhu et al., 2020; Heumann et al., 2021; McKenzie, 2020) are required. Built environments represent the different types of physical features of the urban landscape (Cervero and Kockelman, 1997), which could be categorized by socio-demographics, spatial relationships, point of interest (POI), and land use. Socio-demographics describes the population characteristics such as age, gender, and household. Spatial relationships are the geographical relationships between different areas, such as distance, contiguity, and street networks. POI and land use indicate the functional realm as small as shops or as large as campuses. These factors contribute to depicting the spatio-temporal relationships of trip occurrences that fulfill people's functional requirements, such as commuting, shopping, and recreation. On the other hand, weather conditions directly impact the willingness to rent E-scooters, while periodic factors such as holidays and weekdays imply different activity patterns.
- *Lack of a more systematic integration of the correlations between various factors and the expression of urban complexity.* In current deep-learning approaches, researchers tend to define the problems associated with correlative spatio-temporal factors in different aspects (Kim et al., 2022; Ham et al., 2021). However, the types of factors are limited and simplified compared with multivariate statistical analysis (Bai and Jiao, 2020; Huo et al., 2021). A few studies (He and Shin, 2020; Xu et al., 2021) further utilized more complex expressions as graph representations in their prediction models to describe the relationship between the trip demand and the built environments rather than traditional deep learning methods. However, their methods of concentrating all features on the edges cannot sufficiently represent the regional characteristics.

Overall, we aim to solve the above problems and summarize the main contributions of this study as follows.

- (1) To mitigate the deterioration of trip demand sparsity and imbalance, we recluster spatial data and implement fusion loss functions with a loss penalty.
- (2) To simulate the relationship between urban complexity and shared E-scooter trip demands, we propose different aspects of learning features, including built environments, weather conditions, and periodic features, with comprehensive graph representations.
- (3) We propose a novel model architecture for a graph neural network based on the diffusion convolutional recurrent neural network (DCRNN) (Li et al., 2017), incorporating the above advantages that better capture spatio-temporal dependencies. We prove that our model is superior for overall performance and each trip demand level compared to other benchmark models.

2. Related work

Although shared E-scooter services have developed rapidly in recent years, related research is relatively rare, especially regarding demand prediction. We review the related works, which can be broadly organized into two categories. The first type, which most of the research involves, is based on multivariate statistical analysis and regression and focuses on the correlation between shared E-scooter usage and complex built environments. By contrast, the second type of research is based on deep learning methods and focuses more on temporal dynamic problems of real-time demand forecasting. We will discuss their contributions and limitations as follows.

- *Multivariate statistical analysis and regression for shared E-scooter usage correlation.* These data-driven approaches collected data from different cities and established correlations to the ridership. They comprehensively compare different factors covered by the scope of built environments, weather conditions, or periodic features. Although different studies in different cities sometimes obtain different results, we can summarize the factors they concern in Table 1. Regarding socio-demographics, population (or population density) is discussed most frequently (Lee et al., 2021b; Bai and Jiao, 2020; Huo et al., 2021; Hosseinzadeh et al., 2021; Jiao and Bai, 2020; Merlin et al., 2021). Lee et al. (2021b) reported that the population factor positively influences the number of generated trips. Jiao and Bai (2020) especially indicated a positive correlation between E-scooter trips and areas with higher population density and higher education. Regarding spatial relationships, studies Bai and Jiao (2020), Caspi et al. (2020), Huo et al. (2021), He and Shin (2020) stated that regions near the city center likely have more shared E-scooter traffic. Huo et al. (2021) indicated that the higher street and intersection density with higher accessibility and connectivity is related to the greater number of shared E-scooter trips. Regarding POI and land use, Hosseinzadeh et al. (2021) indicated that commercial and industrial land use affects the shared E-scooter ridership positively and negatively, respectively. Bai and Jiao (2020), Merlin et al. (2021), and Huo et al. (2021) discovered that transit density boosts shared E-scooter ridership because of the first- and last-mile connections. Huo et al. (2021) and Caspi et al. (2020) stated that neighborhoods around university campuses demand a large amount of shared E-scooters. On the other hand, the impact of

Table 1
Summary of the possible factors correlating to shared E-scooter usage in the references.

Categories	Factors	Reference
Socio-demographics	Population	Lee et al. (2021b), Bai and Jiao (2020), Huo et al. (2021), Hosseinzadeh et al. (2021), Jiao and Bai (2020), Merlin et al. (2021)
Spatial relationships	Street and intersection	Caspi et al. (2020), Hosseinzadeh et al. (2021), Huo et al. (2021), Jiao and Bai (2020)
	Distance to city center	Bai and Jiao (2020), Caspi et al. (2020), Huo et al. (2021), Jiao and Bai (2020), Merlin et al. (2021)
POI & Land use	Specific land use	Huo et al. (2021), Jiao and Bai (2020), Lee et al. (2021a), Merlin et al. (2021), Heumann et al. (2021), Hosseinzadeh et al. (2021), McKenzie (2020)
	Land-use entropy	Bai and Jiao (2020), Caspi et al. (2020), Hosseinzadeh et al. (2021), Huo et al. (2021), Jiao and Bai (2020)
Weather	Precipitation, snowfall, temperature	Mathew et al., Zou et al. (2020), Zhu et al. (2020), Noland (2019)
Periodic	Holiday, weekend	Zhu et al. (2020), Heumann et al. (2021), McKenzie (2020)

weather and periodic features on trip demand was also discussed. The studies Mathew et al., Zou et al. (2020), Zhu et al. (2020), Noland (2019) demonstrated that the number of trips decreases rapidly during winter due to the cold and snowfall. The low surface temperature and precipitation also reduce trip frequency. The studies Zhu et al. (2020), Heumann et al. (2021), McKenzie (2020) demonstrated that weekdays, weekends, and holidays exhibit different shared E-scooter trip patterns. Although the above studies comprehensively investigated different possible correlative factors, they did not offer real-time and accurate predictions for each local subarea, which is crucial for service operators. To achieve such precise predictions, deep learning models are being considered as a feasible approach.

- *Deep learning methods for shared E-scooter trip demand prediction.* Traffic demand prediction is one of the fundamental tasks in the transportation domain, and the data-driven deep learning approach with promising result performance has drawn significant attention in recent years. Lv et al. (2014) utilized a stacked autoencoder model (Vincent et al., 2010) to extract traffic flow data. The studies Wu and Tan (2016), Cao et al. (2017), Yu et al. (2017) demonstrated that characterizing both spatial dependency (such as road networks) and temporal dependency (such as traffic congestion) is crucial for traffic prediction, and the integration of recurrent neural network (RNN) (Rumelhart et al., 1985) and convolutional neural network (CNN) (LeCun et al., 1998) models was considered. Zhao et al. (2019) proposed a T-GCN model to replace CNN with a graph convolutional network (GCN) (Kipf and Welling, 2016) to improve the representation of traffic networks with complex topological structures. Furthermore, a DCRNN model (Li et al., 2017) focused on improving long-term prediction (1 h ahead) with nonstationarity caused by recurring events or accidents. Building upon the contributions of prior research, demand predictions for various modes of transportation have been developed. Regarding shared mobility, Lu et al. (2021) utilized a model integrated with CNN and long short-term memory (LSTM, variants of RNN model) (Hochreiter and Schmidhuber, 1997) models to tackle ride-hailing demand prediction by converting demand information as images while Ke et al. (2021) improved with a multi-graph convolutional model rather than a CNN model to address region-based prediction and origin–destination (OD) relations. Xiao et al. (2023) and Xiao et al. (2021) focused on the demand prediction of station-based shared bikes and utilized a gated graph convolutional model to encode spatio-temporal dependencies. Yang et al. (2023) and Li et al. (2023) proposed a model integrated with convolutional and LSTM layers to predict the dockless shared bikes effectively. Compared with traffic prediction based on street segments, it should be noted that these studies (Ke et al., 2021; Yang et al., 2023; Li et al., 2023) focused on the demand prediction of shared mobility modes based on OD regions. To be more precise, the connectivity of street segments remains constant in Euclidean space, allowing the model to directly predict each future traffic flow based on their previous situation and stable in-between interrelationships. However, OD connections are temporally varied based on each trip (i.e., some OD connections may have the origin and destination adjacent, while others may be far apart), which leads to the OD-based prediction being more challenging because of their dynamic interrelationship. As a result, Ke et al. (2021), Yang et al. (2023), and Li et al. (2023) addressed the importance of including additional features related to built environments or weather conditions to improve the prediction capability. It is effective because OD trips are generated to fulfill human needs, and the distinctive characteristics of the built environments in different regions likely influence these behaviors. As an emerging shared

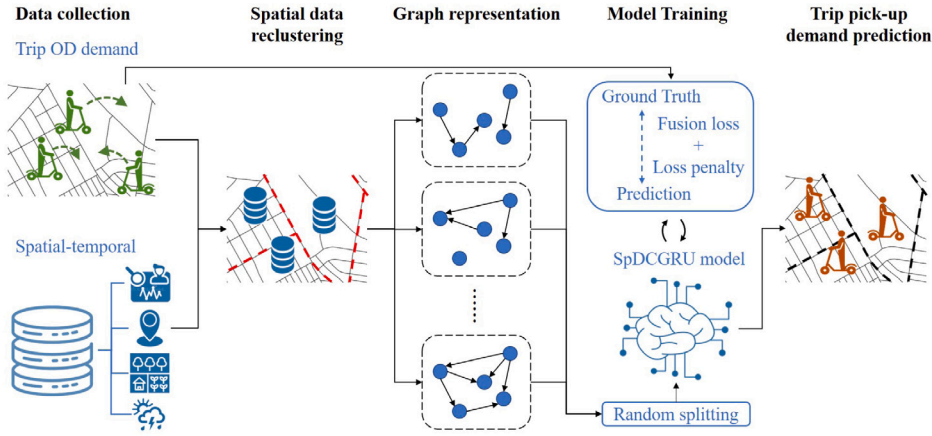


Fig. 1. The framework of shared E-scooter pick-up demand prediction.

mobility mode, shared E-scooters share similarities and differences with ride-hailing and shared bikes. For instance, shared E-scooters and bikes are short-term services and are more susceptible to adverse weather conditions, leading to a significant decrease in usage compared to taxis. Shared E-scooter and ride-hailing services are station-free. Real-time E-scooter demand prediction is commonly performed on an hourly or daily scale (Kim et al., 2022; Ham et al., 2021; Phithakkitnukoon et al., 2021; He and Shin, 2020; Xu et al., 2021). This is because service providers must deploy staff and trucks to manually relocate scooters, balancing supply and demand. However, ride-hailing services require shorter time (10 min or longer) predictions (Ke et al., 2021; Lu et al., 2021) to facilitate the operators in instructing the nearby vehicles to high-demand regions.

To specifically tackle real-time shared E-scooter trip demand prediction, a few researchers adopted deep learning approaches (Kim et al., 2022; Ham et al., 2021; He and Shin, 2020; Xu et al., 2021) and demonstrated the superior performance of their models compared with the classical machine learning methods and regression models. Kim et al. (2022) proposed a reclustering method based on the degrees of each trip connection (Girvan and Newman, 2002) to improve the problem of spatial sparsity and utilized an LSTM model to capture temporal dependency. Ham et al. (2021) focused on densely populated campus neighborhoods and proposed a model architecture constructed by convolutional and RNN-based layers. They demonstrated that LSTM and gated recurrent unit (GRU) (Cho et al., 2014) models shared a similar performance. He and Shin (2020) considered the spatio-temporal features, including the distance between OD regions, POI, connectivity, weather conditions, and days of the week. They utilized graph convolutional layers and capsule routing (Sabour et al., 2017) to capture spatial and temporal dependencies, respectively. Xu et al. (2021) utilized a model combining the GCN and GRU layers and focused on capturing more related features, including socio-demographic, POI functionality, spatial relationships, and weather conditions; however, the built environment factors were all simplified into similarity adjacency matrices with binary labels (i.e., zero or one) to estimate the relationship between the origins and destinations.

Based on previous research, it can be observed that the way of discussing trip demand and usage for shared E-scooters has shifted gradually from statistical methods to more sophisticated spatio-temporal deep learning methods. However, the issue of imbalanced and sparse trip data has not been taken into account properly. For the deep learning models, the presence of extremely imbalanced data is highly likely to cause prediction models to excessively prioritize majority values while overlooking minority values (Krawczyk, 2016; Lin et al., 2017). In other words, accurately predicting minority high-demand values becomes more challenging. Moreover, the additional learning features in the existing deep learning methods were simplified compared to multivariate statistical methods, which is inconsistent with the complexity and uncertainty of shared E-scooter trip demands. There is also a need for a more precise discussion of each feature contribution. In this research, we attempt to fill the research gaps with more extensive period and geographical scope demand data by using more robust models, comprehensive spatio-temporal features, and strategies to improve the prediction performance of sparse and imbalanced demands.

3. Methodology

3.1. Framework of trip demand prediction

Overall, we propose a framework of the trip demand prediction in Fig. 1 and summarize it as follows.

- (1) **Data collection.** Data include OD shared E-scooter trip data, population density, spatial relationships, POI, land use, and weather conditions in Louisville, which will be described in Section 3.5. To improve the learning ability of the prediction model from the data, we preliminarily carry out several statistical analyses as domain knowledge to assist in setting experiments and training strategies accordingly.

- (2) *Spatial data reclustering.* We propose a reclustering method and reorganize the data by mapping them to the geographical divisions discussed in Section 3.6.
- (3) *Graph representation.* The above data are converted into comprehensive graph representations with graphs, nodes, and edges to capture the spatio-temporal dependencies for trip demand prediction. The graph representation is defined in Section 3.3.
- (4) *Model training.* We specifically propose a model for the sparse trip demand prediction for the shared E-scooter service, discussed in Section 4. When training the model, we utilize some strategies regarding the loss functions to obtain more accurate predictions in the variety of trip demand levels discussed in Section 3.6.
- (5) *Pick-up demand prediction.* After training the model with fine-tuning until convergence, we obtain a pre-trained model with optimal performance to predict the pick-up demand in the next hour in each region.

3.2. Problem definition

We define the problems based on a set of spatio-temporal graphs to predict the real-time shared E-scooter trip demands promisingly. Each graph contains different dynamic signals as different trips from the origins to the destinations.

1. **Definition 1:** A graph set $\mathcal{G}$ is a long-term time series of configurations to describe the shared E-scooter OD trip demand with spatial and temporal relationships between regions over a total of T time intervals. It contains dynamic graphs G_t with dynamic temporal signals in each time interval $t \in (0, 1, \dots, T-1)$, which can be represented as $G_t = (V, \mathcal{E}_t, u_t) \in \mathcal{G}$. Here, each graph G_t contains universal features $u_t \in \mathbb{R}^{D_u}$, including periodic and weather features, where D_u represents the dimensions of the universal features. Each graph G_t also contains a node set V and an edge set $\mathcal{E}_t$ defined as follows.
2. **Definition 2:** A node set V represents N divided regions that constitute the overall shared E-scooter service coverage in Louisville. Each region consists of time-varying node features $X_t \in \mathbb{R}^{N \times D_x}$ in each time interval t , where D_x represents the total dimensions of node features. The first dimension of the node feature is the pick-up demand, and the second is the drop-off demand. Others are regional attributes defined by the spatial distance to the city center, street density and intersection density, and population density. Note that only the first and second node features vary over time intervals; others are invariant over time intervals.
3. **Definition 3:** A dynamic edge set $\mathcal{E}_t$ represents all the dynamic trip relationships between the origins and the destinations in each time interval t . Each edge contains edge features $E_t \in \mathbb{R}^{N_{e,t} \times D_e}$ in each time interval t , where $N_{e,t}$ represents the total number of edges varying in each time interval t and D_e represents the total dimensions of edge features. The spatial and built environment relationships between the trip origins and the destinations, such as region center distance, contiguity, POI similarity (Tang et al., 2021; Xu et al., 2021), and difference of land-use entropy (Bai and Jiao, 2020; Caspi et al., 2020; Hosseinzadeh et al., 2021; Huo et al., 2021; Jiao and Bai, 2020), define the edge features. Regarding the edge indices, $Ein_t \in \mathbb{R}^{2 \times N_{e,t}}$ represents the sparse edge index matrix in time interval t , identifying the regions which exhibit OD trips.
4. **Problem:** Based on the above graph, node, and edge definitions, we define a function f that maps the graphs $G_{(t-p+1):t}$ and the node features $X_{(t-p+1):t}$ for the previous series of time intervals from $(t-p+1)$ to t , to the node features X_{t+1} in the next interval $(t+1)$, as given by Eq. (1):

$$X_{t+1} = f(G_{(t-p+1):t}, X_{(t-p+1):t}) \quad (1)$$

Note that in practice, we focus on predicting the pick-up demands, i.e., the first dimension of node feature x_{t+1}^0 , which can be defined as the target of $y_{t+1} \in \mathbb{R}^N$. The model aims to minimize the loss between the target of y_{t+1} and the prediction of $\hat{y}_{t+1} \in \mathbb{R}^N$.

3.3. Graph representation and built environment factors

Based on the correlation research in Section 2, we define our comprehensive spatio-temporal features, including built environments, weather conditions, and periodic features. Based on each feature's characteristic, their learning information is embedded into graphs, nodes, and edges, as detailed below.

- *Universal features.* They consist of weather conditions and periodic features. We apply time-series information, including year, month, week, and hour, as features and transform them into cyclical representations via cosine and sine functions based on their periodic behaviors (see Supplementary Materials 1).
- *Node features.* We extract the hourly pick-up demand as the first dimension feature, which is specified with negative integers to distinguish it from the drop-off demand as positive integers; by contrast, the hourly trip drop-off demand is the second dimension feature with positive integers. The sum of hourly trip pick-up and drop-off demands is the third dimension feature, which would be zero if rented and returned E-scooters share the same numbers. Other features representing urban built environments include spatial distance to the city center, street density, intersection density, and population density for each region.
- *Edge features.* The three spatial relationships between the origins and the destinations are distance, POI similarity (Tang et al., 2021; Xu et al., 2021), and difference in land-use entropy (Bai and Jiao, 2020; Caspi et al., 2020; Hosseinzadeh et al., 2021; Huo et al., 2021; Jiao and Bai, 2020). We calculate POI similarity using Pearson's correlation coefficient to estimate the functional similarity between two regions. Moreover, we utilize land-use entropy ranging from 0 to 1 to estimate the equilibrium between

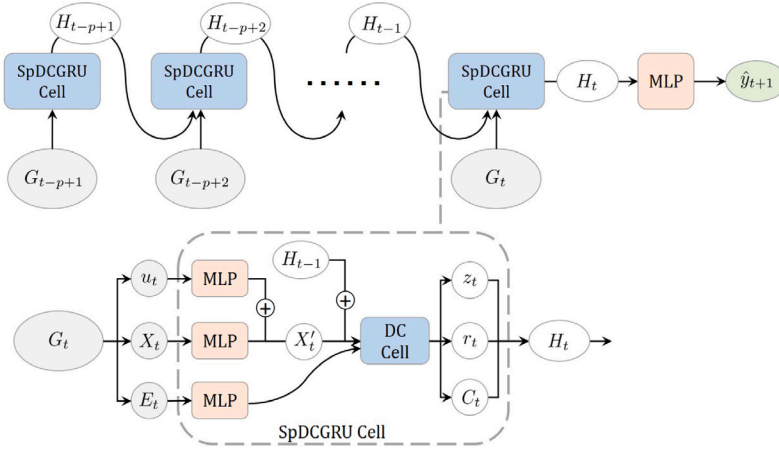


Fig. 2. Proposed model architecture, consisting of SpDCGRU cells to capture spatial and temporal dependencies.

all available land uses for each region and calculate the differences between origins and destinations. See Supplementary Materials 1 for detailed definitions of POI similarity and land-use entropy. Note that if the convolutional model is GCN-related, then edge features will be considered a feature-concatenated weight matrix in response to the adjacency matrix for edge connections.

3.4. Model architecture

We introduce a modified version of DCRNN called Sparse Diffusion Convolutional Gated Recurrent Unit (SpDCGRU) as a graph deep learning framework tailored specifically to extract complex urban features and forecast imbalanced and sparse shared E-scooter trip demand.

The overall model architecture is shown in Fig. 2. Its major structure is derived from the model of a GRU (see Supplementary Materials 3), which has been demonstrated to successfully capture the temporal dependency (Ham et al., 2021; Xu et al., 2021). Overall, to obtain the prediction $\hat{y}_{t+1}$ compared with the target y_{t+1} in the next $(t + 1)$ time interval, the inputs are the previous p numbers of graphs from G_{t-p+1} to G_t . Through the iterative process of the SpDCGRU cell, similar to the GRU model, each step generates each hidden state from H_{t-p+1} to H_t ; it is then passed to the next SpDCGRU cell in the next step until obtaining H_t . After this sequential encoding process, we utilize a multilayer perceptron (MLP) to decode the hidden state H_t and finally compute the node-wise predictions $\hat{y}_{t+1}$ (i.e., the pick-up demand prediction for the next time interval for each region).

To better capture the spatial dependency, we utilize advanced convolutional layers called diffusion convolution (DC) layer (Li et al., 2017) rather than other typical graph neural networks, such as GCN (Kipf and Welling, 2016), GAT (Veličković et al., 2017), and GIN (Xu et al., 2018). The DC layer learns the concept of random walks in message passing on the graph with a restart probability $\alpha \in [0, 1]$ and a state transition matrix $D_O^{-1}W$. Here, $W \in \mathbb{R}^{N \times N \times D_e}$ is a weighted adjacency matrix representing edge features, and D_O represents the degree matrix of W in the forward direction. After sufficiently long time steps, the Markov process converges into a stationary distribution $P \in \mathbb{R}^{N \times N}$, which can be calculated in a closed form as in Eq. (2) (where i th row of $P_{i,:} \in \mathbb{R}^N$ represents the likelihood of diffusion from node $v_i \in V$):

$$P = \sum_{k=0}^{\infty} \alpha(1 - \alpha)^k (D_O^{-1}W)^k \quad (2)$$

where k is the diffusion step. In practice, k is set to a finite number for the diffusion process with a trainable weight for each step. Considering the destinations likely influence the trips (i.e., citizens fulfill their functional purposes in the destinations), we also similarly include the reversed direction of the diffusion process with different trainable weights. Such a bidirectional diffusion process, defined by Eq. (3), can provide more complexity and flexibility in capturing spatial dependency:

$$H' = \sigma \left(\sum_{k=0}^{K-1} (\theta_{k,1} (D_O^{-1}W)^k + \theta_{k,2} (D_I^{-1}W^T)^k) H \right) \quad (3)$$

where H represents the hidden state, and H' is the outcome from the diffusion convolution filters; $\sigma(\cdot)$ represents the nonlinear activation function; θ represents the parameters for the diffusion weights; and $D_O^{-1}W$ and $D_I^{-1}W^T$ represent the transition matrices of the forward and reversed diffusion processes, respectively.

To capture spatial and temporal dependencies simultaneously, the SpDCGRU cell merges the DC cell into the GRU model rather than constructing the convolution and recurrent layers sequentially (Ham et al., 2021; Xu et al., 2021). First, in each SpDCGRU cell in the time interval t , the input of graph G_t , containing universal features u_t , node features X_t , and edge features E_t , is encoded by

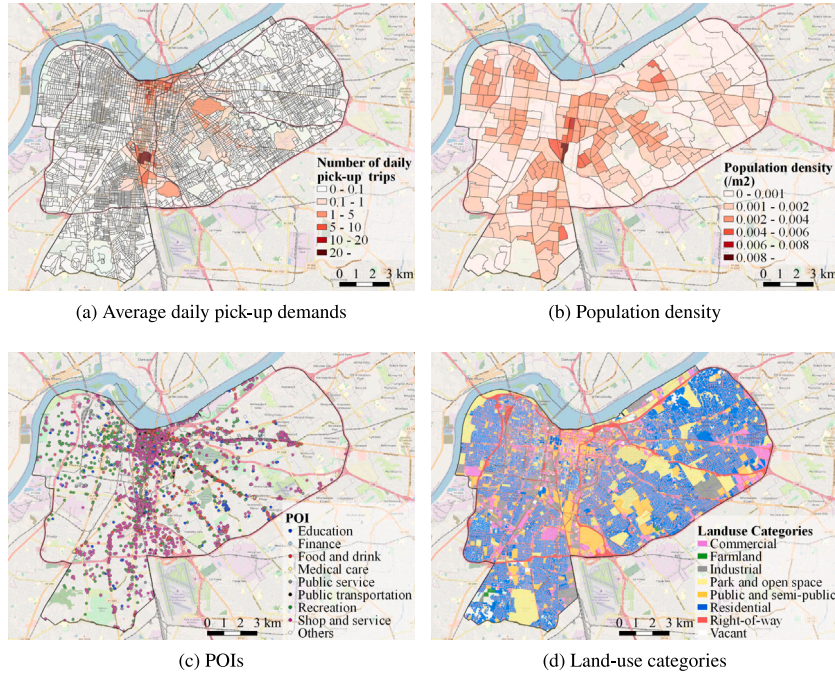


Fig. 3. Shared E-scooter trip demand and built-environment distributions in Louisville.

the MLP individually. Second, we convert the encoded universal features into the size of nodes and subsequently concatenate it with the encoded node features as the collaborative node features X'_t ; this is defined by Eq. (4). Third, we utilize the collaborative node features X'_t , the previous outcome of the hidden state H_{t-1} , and the encoded edge features as inputs to process the DC layer and obtain the reset gate r_t , update gate z_t , the hidden state for storing current memory contents C_t , which can consequently outcome the hidden state H_t by Eq. (8) for the overall long-term and short-term memory contents. It should be noted that we replace the GRU's original matrix linear multiplications with the DC layer and utilize the concatenation operator with X'_t and H_{t-1} instead of the aggregation operator to provide more complexity and flexibility for capturing spatio-temporal dependencies defined by from Eq. (5) to Eq. (8):

$$X'_t = [MLP_u(u_t) \parallel MLP_X(X_t)] \quad (4)$$

$$r_t = \sigma(\Theta^r \star_G [X'_t \parallel H_{t-1}] + b^r) \quad (5)$$

$$z_t = \sigma(\Theta^z \star_G [X'_t \parallel H_{t-1}] + b^z) \quad (6)$$

$$C_t = \tanh(\Theta^C \star_G [X'_t \parallel (r_t \odot H_{t-1})] + b^C) \quad (7)$$

$$H_t = z_t \odot H_{t-1} + (1 - z_t) \odot C_t \quad (8)$$

where $[\parallel]$, $\sigma(\cdot)$, $\tanh(\cdot)$, and $\odot$ represent the concatenation operator, the nonlinear sigmoid activation function, the hyperbolic tangent function, and the Hadamard product (element-wise multiplying operator), respectively; $\star_G$ represents the diffusion convolution filters; Θ^r , Θ^z , and Θ^C represent the learning parameters for the corresponding gates or states, while b^r , b^z , and b^C represent the bias for the corresponding gates or states.

3.5. Data description

We collected data from a shared E-scooter pilot program implemented in Louisville, Kentucky, USA. The other data sets regarding spatio-temporal features corresponding to the same geographical and period scopes as the trip data are described as follows. In this research, we focus on the next hourly pick-up demands as our prediction targets: the number of shared E-scooters rented from each region in the next hour.

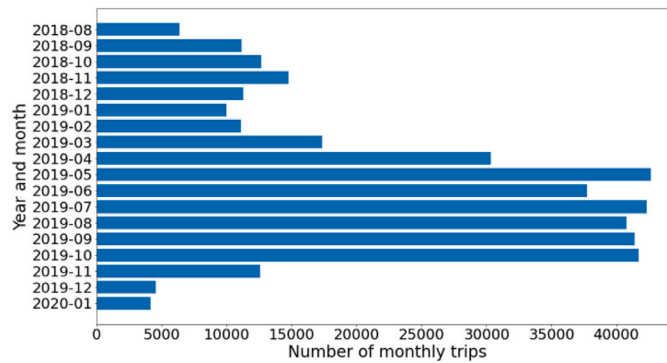


Fig. 4. Aggregated trip demand for each month.

Table 2

Definition of each multi-classification label corresponding to each pick-up demand and level.

Label	0	1	2	3	4	5
Trip demand	0	1	2	3–4	5–10	11–
Trip demand level	Zero			Low		High
Proportion (%)	79.0			19.9		1.1

- *Shared E-scooter OD trip data set.* The data set, provided by the [Office of Civil Innovation and Technology, Louisville-Jefferson County Metro Government \(2021\)](#), includes 505,993 trips from August 9, 2018, to January 31, 2020 (541 days), which are then filtered to 405,147 trips, with an average of approximately 750 trips per day. In [Fig. 3\(a\)](#), we configure the pick-up demand distribution by regions of census blocks with 5,122 blocks; the block divisions are provided by the United States Census Bureau (2020b). The regions in darker red have a greater number of average daily pick-up trips. Note that the trip demand significantly varies from region to region and is concentrated in downtown, university campuses, and parks around the middle of the city. By contrast, most blocks present significantly lower trip demands with no more than 0.1 during a day; in other words, approximately zero pick-up demand in each hour. In addition, as the trip demands also highly vary over months in [Fig. 4](#), the one from April to October 2019 is nearly double that of the remaining months, exacerbating the issues of imbalance and sparsity. To overcome this, we propose periodic features in [Section 3.3](#) and strategies for the imbalanced data in [Section 3.6](#).
- *Population data set.* The data set for each census block group is collected from the [United States Census Bureau \(2020a\)](#). In [Fig. 3\(b\)](#), we visualized the population density across different block groups, with the darker red areas representing the higher values. Population density is defined as the population volume divided by the area (m^2) it belongs to. It can be seen that the distribution of population density is concentrated in several areas, yet it does not exhibit an apparent correlation with the distribution of pick-up demand. However, we still consider it one of our model learning features based on the research ([Lee et al., 2021b](#); [Bai and Jiao, 2020](#); [Huo et al., 2021](#); [Hosseinzadeh et al., 2021](#); [Jiao and Bai, 2020](#); [Merlin et al., 2021](#)).
- *Street network and POI data set.* These two data sets are collected from [OpenStreetMap \(2022\)](#), an open geographic database updated and maintained by volunteer collaboration. It contains whole street networks in Louisville with polyline geometries and intersections, as well as POIs with their locations and amenity types. We re-categorized the POI types as recreation, public services, shops and private services, food and drink, finance, public transportation, education, medical care, and others. Their distribution is shown in [Fig. 3\(c\)](#). Most POIs exist downtown and around the university campus, generally consistent with the trip demand pattern.
- *Land-use data set.* The data set is collected from the [Office of Civil Innovation and Technology, Louisville-Jefferson County Metro Government \(2020\)](#). Land uses in Louisville are categorized into commercial, farmland, industrial, park and open space, public and semi-public, residential, right-of-way, and vacant in each census block. We visualized the land-use distribution in [Fig. 3\(d\)](#). Most land is covered by residential usage, while the downtown, with the small and compressed census blocks in the middle and upper side of the city, contains the most complex commercial behavior and higher number of trips.
- *Weather data set.* The data set with daily reviews is collected from the [National Centers for Environmental Information, NOAA \(2020\)](#). We utilized factors including precipitation, snowfall, and average temperature during the day to present the overall weather conditions in Louisville, affecting the travelers' behavior significantly.

3.6. Strategies for the imbalanced data

- *Spatial data reclustering.* To integrate different learning features from different data sets measured from different standard divisions (such as census blocks and census block groups), the data information must be reaggregated, redistributed, and

re-mapped into the same geographical divisions. We propose reclustering the prediction divisions from geographical census blocks to reclustered regions to tackle the extreme proportion of the zero trip demand in some areas. Instead of a regular grid of divisions, the block-based heterogeneous and administrative domain divisions likely better reflect the relationship between the shared E-scooter trips and the built environment factors (Hosseinizadeh et al., 2021; Ke et al., 2021). We utilize a hierarchical cluster method (Murtagh and Contreras, 2012) by extracting the pick-up demand as the variable and restraining the clustering process followed by rook contiguity (i.e., only the block neighbors are clustered as a new region). The result of the reclustered regions is shown in Fig. 5. The deeper red color represents higher pick-up demands. The model training will neglect the two white reclustered regions on the right and left sides since shared E-scooter trips barely occurred during the historical period. Consequently, only 43 reclustered regions remain to be processed in the following experiments.

- **Fusion loss function and evaluation metrics.** We propose a fusion loss function incorporating the mean absolute error (MAE) (Hyndman and Koehler, 2006) and the multi-classification loss with a loss penalty to tackle the extremely sparse and imbalanced data. The multi-classification problem is constructed by categorizing the original regression targets as different trip demand levels to improve each level's performance and address the minority of higher demand. In Fig. 6, we aggregate the status for the number of hourly pick-up e-scooters in each region to measure the imbalanced demands. It can be seen that zero trip demand accounts for approximately 79.0%, while one trip demand accounts for approximately 10.6%, and others are even fewer. Based on these results, we take advantage of Jenks natural breaks algorithm (North, 2009) to categorize and define the multi-classification label in Table 2, with six classes. For practical usage of the reallocation strategies, we further simplify them as three levels of renting demands, including zero, low, and high, which correspond to the pick-up demands of 0, from 1 to 2, from 3 to 10, and larger than 10, respectively. Accordingly, we define evaluation metrics to evaluate the overall performance, including the MAE, root mean square error (RMSE) (Hyndman and Koehler, 2006), and F1 score (Powers, 2020). The MAE intuitively indicates the results of the difference in the number of trip demands and corresponding predictions (for instance, an MAE of 1 signifies a difference of 1 E-scooter between the actual demand and the predicted value), while the RMSE can highlight the outliers by using square root distance of them. In addition, we utilize the F1 score to evaluate the multi-classification overall performance. In this study, we prefer not to utilize the mean absolute percentage error (MAPE) (Hyndman and Koehler, 2006) because trip demands exhibit extreme sparsity and imbalance, leading to MAPE's failure when encountering zero or close-to-zero ground truth. In addition, we utilize the area under the curve (AUC) of the receiver operator characteristic (ROC) (Powers, 2020) to highlight the metrics of each trip demand level, especially for non-zero demands. See Supplementary Materials 2 for more comprehensive definitions of those evaluation metrics, including the MAE, RMSE, F1 score, and AUC. Essentially, tackling the imbalanced multi-classification, the focal loss is suggested by Lin et al. (2017) as Eq. (9). It down-weights the contribution of majority samples and emphasizes the contribution of minority samples based on the possibility distribution of occurrence.

$$Focal\ loss = -(1 - \hat{p}_y)^\gamma \log(\hat{p}_y) \quad (9)$$

In the above, $\{\hat{p}_0, \dots, \hat{p}_y, \dots, \hat{p}_{K-1}\} \in [0, 1]^K$ represents a set of the estimated probability distribution over K classes. $\gamma \geq 0$ is a tunable focusing parameter, representing how much higher-confidence correct predictions contribute to the overall loss; for instance, the common multi-class softmax cross-entropy loss is recovered by setting $\gamma = 0$. In this research, we have fine-tuned the training hyperparameters γ as 2. K is set as 6 based on Table 2. In addition, We define the fusion loss function to incorporate with the MAE loss and the focal loss multiplied by the weighting parameter (w_{focal}), which is a training hyperparameter set as 0.5. To avoid the pre-trained model that tends to erroneously predict all demands as zero because of the high imbalance of zero trip demand in the ground truth, a loss penalty function $f_{penalty}$ is executed to amplify the original loss with a weighting parameter of penalty when the ground truth is not equal to zero, and the prediction is. In this research, we have fine-tuned the weighting parameter of the penalty as 5. Finally, The fusion loss is defined by Eq. (10):

$$Fusion\ loss = f_{penalty} (MAE + w_{focal} \times Focal\ loss) \quad (10)$$

4. Experimental results and discussion

Regarding the experiment settings, we utilized the fine-tuned previous 6-hour historical data to predict the next one discussed in Section 5. The learning rate was set to 10^{-5} . The data set was randomly split into training and validation data sets with a ratio of 0.9. The number of hidden units was 128. The model was trained using the Adam optimizer, and the training epoch was 300 with early stopping. Each test was trained repetitively five times to obtain average evaluation metrics. In the following sections, we will demonstrate the performance compared to other research, the contribution of different learning features, and the contribution of fusion loss.

4.1. Model comparison

The benchmark models with state-of-the-art graph neural networks and variants of our models are described as follows.

- **GCRN_LSTM.** The graph convolutional recurrent network (GCRN) (Seo et al.) was proposed to predict structured and sequential data based on the LSTM model. It replaces the original linear matrix multipliers with graph convolution layers in each gate of the GRU model to capture spatio-temporal dependencies.

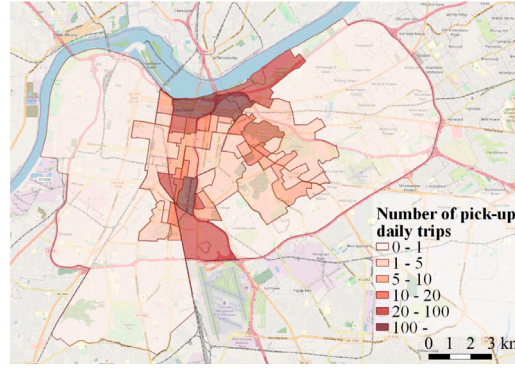


Fig. 5. Average daily pick-up demands after reclustering.

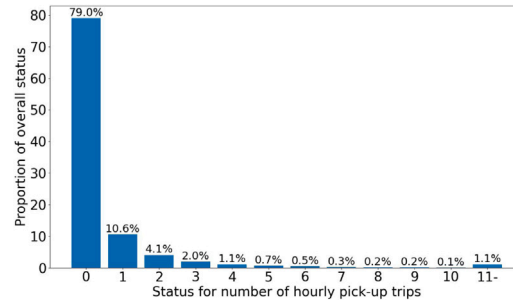


Fig. 6. Aggregating the status of hourly pick-up demands by regions and periods.

- *GCRN_GRU*. A variant model from GCRN replaced the LSTM model with the GRU model.
- *T-GCN*. The temporal graph convolutional network (Zhao et al., 2019) was introduced specifically for traffic forecasting, which sequentially processed the graph convolutional layer first, followed by the GRU model. It also utilized the concatenation operator to replace the summation operator of X_t and H_{t-1} in the GRU model.
- *A3T-GCN*. The attention temporal graph convolutional network (Bai et al., 2021) was also specifically proposed for traffic forecasting, and it utilized the attention mechanism to capture spatial dependency.
- *SpGRU+GAT*. The variant of our model replaces DC layers with GAT layers (See Supplementary Materials 3).
- *SpGRU+GIN*. The variant of our model replaces DC layers with GIN layers (See Supplementary Materials 3).
- *SpGRU+GCN*. The variant of our model replaces DC layers with GCN layers (See Supplementary Materials 3).

We present the results in Table 3, where the bold text indicates the best performance for each metric. Our SpDCGRU model is generally superior to other models in terms of the MAE, RMSE, and F1 score for overall performance and AUC metrics for each trip demand level. Compared to GCRN_GRU, our model shows improvements of 4.75%, 3.26%, 0.39%, 1.39%, 1.69%, 0.70%, 1.63%, 2.56%, and 1.23%, respectively. The results reveal that our model's diffusion mechanism with separate forward and backward message-passing can more effectively simulate the trip demands with extreme spatio-temporal sparsity and uncertainty. Regarding AUC metrics, our model achieves better results for AUC_0 and AUC_5 compared to other AUC metrics, indicating that the model is adept at predicting the majority class with the zero trip demand level and the minority class with the high trip demand level. By contrast, predicting the low trip demand level with the number of E-scooters ranging from 1 to 10 is relatively challenging, especially for the interval from 2 to 4 (represented by AUC_2 and AUC_3).

A comparison of the performance of other models for the prediction task indicates the following. Compared with GCRN_LSTM and GCRN_GRU, the GRU-based model performs slightly better for all overall performance metrics, with an improvement of 0.6% for the MAE and less than 28% training time. It confirms the advantage of constructing our model based on GRU. Furthermore, the model structure of merging the convolution layer into the GRU model is more favorable than sequentially constructing them. It is confirmed by performance improvements of 18.44%, 11.61%, 5.75%, 4.24%, 4.97%, 6.23%, and 14.51% for the MAE and each AUC metric, respectively, in the comparison between GCRN_GRU and T-GCN. Surprisingly, A3T-GCN utilized the attention mechanism, likely failing this imbalanced demand prediction task for each AUC metric.

Regarding our variants of models, we compare the performances of different convolution layers, namely GAT, GIN, GCN, and DC. The GIN-based model (SpGRU+GIN) performs the worst on each AUC metric; as a result, it is unsuitable for tackling the extreme imbalance of zero demands since its concept of updating the summation messages likely worsens the effects of imbalanced data. Notably, SpGRU+GCN and SpDCGRU are the best two models in the experiments; furthermore, the DC layer with the diffusion

Table 3
Model performance for the shared E-scooter trip demand prediction task.

Method	Overall performance			Trip demand level					
				Zero	Low		High		
	MAE	RMSE	F1	AUC ₀	AUC ₁	AUC ₂	AUC ₃	AUC ₄	AUC ₅
GCRN_LSTM	0.444	1.413	0.774	0.769	0.628	0.562	0.591	0.645	0.865
GCRN_GRU	0.441	1.364	0.774	0.778	0.638	0.566	0.603	0.648	0.882
T-GCN	0.522	2.033	0.739	0.688	0.601	0.542	0.573	0.607	0.754
A3T-GCN	0.918	3.314	0.702	0.500	0.500	0.500	0.500	0.500	0.500
SpGRU+GAT	0.476	1.547	0.766	0.752	0.607	0.561	0.587	0.651	0.871
SpGRU+GIN	0.705	3.327	0.697	0.622	0.575	0.510	0.504	0.501	0.500
SpGRU+GCN	0.434	1.337	0.777	0.786	0.647	0.569	0.611	0.680	0.878
SpDCGRU	0.421	1.321	0.777	0.789	0.649	0.570	0.613	0.665	0.893

mechanism is an advanced graph convolution layer with more complexity for predicting uncertain trip demands. Compared to SpGRU+GCN, SpDCGRU generally demonstrates superior prediction performance, especially for the improvement of 3.09% MAE, 1.21% RMSE, and 1.68% high trip demand level.

4.2. Average prediction performance over different periods

We further discuss the average prediction performance over different periods corresponding to their average trip demands in Fig. 7. Regarding the hourly behaviors in Fig. 7(a), it reveals that the average number of pick-up trips starts to increase from midnight, reaching its peak at 3 P.M., and then gradually declines until midnight again. This significant difference in the number of trips over hours causes the prediction model to exhibit a relatively inferior performance of the F1 score with about 0.65 from 11 A.M. to 8 P.M. in Fig. 7(c). By contrast, the other average hourly performance of the F1 score is relatively higher, from about 0.7 to 0.9. From 11 A.M. to 8 P.M., due to the higher number of users, there is also greater variation in travel behavior across different regions, making predictions more challenging. Other related studies have observed similar results (Kim et al., 2022; Phithakkitnukoon et al., 2021; Xu et al., 2021). If detailed background information about each journey's user (such as age, gender, etc.) can be obtained along with the corresponding regions' land use and POI data in future studies, it should enhance the model's performance. On the other hand, regarding the daily behaviors in Fig. 7(b), it can be seen that Saturday shows the most substantial frequency of average E-scooter rental usage, followed by Friday. The usage frequency on other days is relatively similar. Compared to our model performance in Fig. 7(d), each average F1 score is approximately consistent daily, with only a slightly diminished performance observed on Saturday because of the largest number of trips. Compared with Figs. 7(c) and 7(d), predicting the hourly demand is likely more complex and uncertain than the daily demands because the daily demands can be more effectively learned from their overall patterns by extracting periodic features.

4.3. Ablation study for the additional learning features

We conduct an ablation study to compare the contributions of the proposed learning features discussed in Section 3.3. The ablation study evaluates the model's performance after systematically removing a specific feature to determine its individual contribution (Meyers et al., 2019). The results are shown in Table 4, where the red text indicates the worst performance for each metric and the highest contribution of the corresponding removed features. The variants of feature inclusion are defined as follows.

- *w/o Population density.* The node feature of population density is removed.
- *w/o Spatial relation.* The part of node features are removed, including the distances to the city center and the streets with intersection density. The edge features are also removed, including the distances between origins and destinations and the region contiguity between origins and destinations.
- *w/o POI.* The edge feature of POI similarity is removed.
- *w/o Land use.* The edge feature of the difference in land-use entropy is removed.
- *w/o Weather.* The universal features regarding the weather conditions, including precipitation, snowfall, and average temperature, are removed.
- *w/o Periodic.* The universal features regarding periodic features, including year, month, week, day, time, and corresponding cyclical representations, are removed.

Our findings indicate that the periodic features make the most significant contribution to the zero trip demand level. This is confirmed by the decrease of the AUC₀, AUC₁, and AUC₂ metrics by 3.17%, 3.54%, and 2.11%, respectively, and overall evaluation metrics of the MAE, RMSE, and F1 score deteriorating by 14.73%, 12.57% and 2.70%, respectively. These results are consistent with the study (He and Shin, 2020) that the feature of the week of a day influences most. The second predominant feature is the weather conditions, with the second highest contribution of 6.65% in the MAE, 9.24% in the RMSE, and 1.16% in the F1 score; its

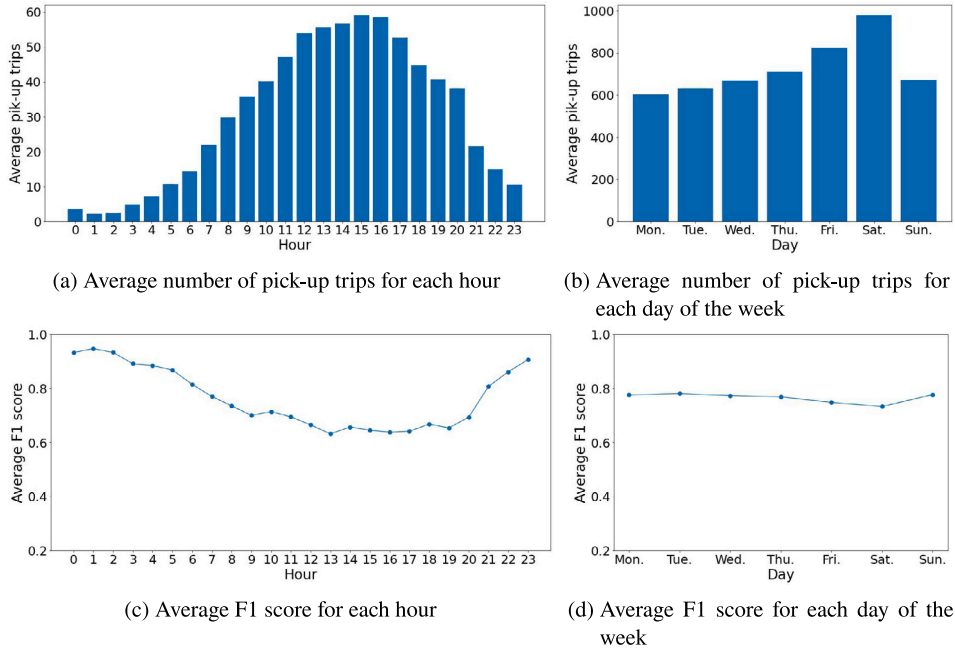


Fig. 7. Comparison of average pick-up demands and the average prediction performance (F1 score) for different periods.

Table 4
Ablation study for the additional learning features.

(w/o) Learning feature	Overall performance			Trip demand level					
				Zero	Low				High
	MAE	RMSE	F1	AUC ₀	AUC ₁	AUC ₂	AUC ₃	AUC ₄	AUC ₅
Population density	0.444	1.373	0.767	0.783	0.644	0.569	0.606	0.655	0.869
Spatial relation	0.438	1.369	0.771	0.771	0.633	0.559	0.592	0.666	0.875
POI	0.439	1.409	0.769	0.775	0.640	0.562	0.603	0.657	0.880
Land use	0.426	1.325	0.773	0.786	0.651	0.566	0.608	0.662	0.866
Weather	0.449	1.443	0.768	0.780	0.646	0.567	0.613	0.660	0.853
Periodic	0.483	1.487	0.756	0.764	0.626	0.558	0.594	0.658	0.891
SpDCGRU	0.421	1.321	0.777	0.789	0.649	0.570	0.613	0.665	0.893

importance is consistent with the research results (Xu et al., 2021). It also declines the most for AUC₅, with 4.48%, representing the high trip demand level. This seems reasonable since the poor weather conditions significantly discourage the intention to rent shared E-scooters. In addition, it is worth noting that the training features of spatial relation and population density play the most crucial role in AUC₃ and AUC₄, with 3.43% and 1.50% contribution, respectively. This result reflects the importance of defining the delicate features in spatial relations, which were oversimplified in previous research. By contrast, the training features of POI and land-use entropy are not relatively significant compared to others; however, both still contribute to the training model. Removing POI features would deteriorate the performance by 4.28%, 6.66%, and 1.03% for the overall MAE, RMSE, and F1 score, respectively, while removing land-use entropy features would deteriorate the performance by 1.19%, 0.30%, and 0.51% for the overall MAE, RMSE, and F1 score, respectively.

4.4. Contribution of fusion loss

We estimate the contribution of our fusion loss strategies defined in Section 3.6. We present the results in Table 5, where the bold and red values indicate each metric's best and worst performances, respectively. The variants of experiments using different loss functions are defined as follows.

- *Single MAE*. It only carries out the MAE loss in the model's backpropagation process and removes the multi-classification loss.
- *Cross-entropy*. It still conducts the fusion loss but utilizes the cross-entropy loss instead of the focal loss for the multi-classification part.
- *w/o Loss penalty*. It removes the loss penalty mechanism from the fusion loss.

Table 5
Performance comparison of different strategies of loss functions.

Loss function	Overall performance			Trip demand level					
				Zero	Low				High
	MAE	RMSE	F1	AUC ₀	AUC ₁	AUC ₂	AUC ₃	AUC ₄	AUC ₅
single MAE	0.429	1.362	0.776	0.784	0.648	0.565	0.612	0.667	0.869
cross-entropy	0.482	1.797	0.774	0.717	0.619	0.547	0.582	0.625	0.791
w/o Loss penalty	0.394	1.423	0.772	0.688	0.542	0.552	0.602	0.666	0.872
SpDCGRU	0.421	1.321	0.777	0.789	0.649	0.570	0.613	0.665	0.893

Table 6
Performance comparison of historical data from different hourly periods utilized as inputs to predict the next-hour trip demands.

Previous hourly data as inputs	Overall performance			Trip demand level					
				Zero	Low				High
	MAE	RMSE	F1	AUC ₀	AUC ₁	AUC ₂	AUC ₃	AUC ₄	AUC ₅
1-hour	0.434	1.328	0.780	0.767	0.618	0.571	0.607	0.655	0.882
3-hour	0.427	1.357	0.777	0.788	0.643	0.572	0.611	0.667	0.888
12-hour	0.422	1.308	0.774	0.786	0.648	0.567	0.615	0.670	0.884
6-hour	0.421	1.321	0.777	0.789	0.649	0.570	0.613	0.665	0.893

Our findings indicate that utilizing the focal loss is more promising than the single MAE loss for most evaluation metrics, including the MAE, RMSE, F1 score, and the AUC metrics (except AUC₄), which show improvement of 1.90%, 3.10%, 0.13%, 0.63%, 0.15%, 0.88%, 0.16%, and 2.69%, respectively. It mainly affects AUC₅ and benefits the prediction of the highest trip demand level, confirming the focal loss theory. By contrast, surprisingly, the fusion loss with the cross-entropy loss generally performs worse. It exhibits a significant setback of 14.49% on the MAE and 36.03% on the RMSE. Repeated experiments also indicated that utilizing cross-entropy loss sometimes leads to the model failing to predict each AUC metric. On the other hand, although the experiment of removing the loss penalty performs better in terms of the MAE, other metrics reveal the opposite results. To be more specific, the F1 score and each AUC metric except AUC₄ respectively decline by 0.64%, 12.80%, 16.49%, 3.16%, 1.79%, and 2.35% compared to SpDCGRU, especially affecting the zero and low trip demand levels. It also exhibits a setback of 7.72% on RMSE. Notably, carrying out the loss penalty likely leads to the model requiring more epochs to converge during the training process since some of the sample losses are amplified until they can be correctly classified as either zero trip demand level.

5. Discussion

To optimize prediction performance using past hourly data, we fine-tuned the hyperparameters pertaining to the data duration, using this information as input to forecast next-hour trip demands. In Table 6, the average results from five repeated experiments are presented. Bold values highlight the best performances for each metric, while red values denote the worst. The average previous 1-hour results are the worst, as the MAE is significantly inferior, and most trip demand level metrics reveal similarly. The 6-hour results outperform others in MAE, the zero trip demand level, and the high trip demand level. Although its performance of the low trip demand level is not the best, the values are close to the best one. The comparison results also reveal that the previous short-term historical data (below 3 h) is insufficient to predict the shared E-scooter demand. Although the average duration of E-scooter usage for all journeys is approximately 16 min, the comparison of prediction results reveals that the model requires previous longer-period trip data to learn the underlying relationships (in association with additional built environments, weather conditions, and periodic features). On the other hand, previous 12-hour data likely contains excessive information to misguide the prediction model according to the performance results. Even if it achieves the lowest RMSE, its AUC metric performance for trip demand levels is not as good as the 6-hour model, making it less suitable for practical use. In addition, training the model on the previous 12-hour data takes an average of 41 h, whereas using the previous 6-hour data takes about 22 h. Considering all factors, we view the previous 6-hour data as the ideal hyperparameter for optimal performance, particularly in relation to the high trip demand level — a minority class that is difficult to forecast.

In addition, we explore the prediction performance between our proposed model and the DCRNN model in Table 7. The 'vanilla DCRNN' refers to using the original DCRNN model without the additional learning features and strategies we introduced for imbalance issues. Conversely, the 'DCRNN with additional features' incorporates those extra learning aspects into the original DCRNN model. The red and bold text indicates each metric's worst and best performances. Broadly, the SpDCGRU model outperforms the others, illustrating that our modifications effectively enhance the DCRNN's predictive capability for shared E-scooter demands, both overall and at each demand level. The performance of vanilla DCRNN is the least satisfactory, indicating the importance of adding additional features. Regarding the DCRNN with the additional features, although its MAE is exemplary, other metrics all fall short. Compared to the SpDCGRU model, each metric of trip demand levels also indicates that the strategies for the imbalanced data we proposed in the SpDCGRU model are necessary.

Table 7
Performance comparison between DCRNN and SpDCGRU.

Model variants	Overall performance			Trip demand level					
				Zero	Low				High
	MAE	RMSE	F1	AUC ₀	AUC ₁	AUC ₂	AUC ₃	AUC ₄	AUC ₅
Vanilla DCRNN	0.457	1.456	0.761	0.749	0.621	0.539	0.571	0.632	0.842
DCRNN with additional features	0.383	1.332	0.774	0.675	0.547	0.550	0.595	0.655	0.871
SpDCGRU	0.421	1.321	0.777	0.789	0.649	0.570	0.613	0.665	0.893

On the other hand, Table 4 illustrates how population density features enhance the predictive capability of our model. In addition, there are still other potential socio-demographic factors that have not been discussed. We have tried to extract the related data, including the sex ratio, median age, median income, and unemployment rate from the American Community Survey (ACS) (United States Census Bureau, 2020b) as socio-demographic features in our learning model. The model's results indicate that the overall evaluation metrics (including the MAE, RMSE, and F1 score) are 0.417, 1.327, and 0.780, respectively. Regarding each trip demand level, the performance of the AUC metrics is 0.783, 0.644, 0.568, 0.605, 0.657, and 0.886, respectively. Compared with SpDCGRU's results, adding these features improves overall performance slightly; however, the metrics of trip demand levels reveal the opposite outcome, which is not conducive to practical e-scooter reallocation applications. The performance of the metrics of trip demand levels does not show improvement, likely due to the inadequate accuracy and granularity of this ACS socio-economic data. First, as a household survey collected from mailed questionnaires, telephone interviews, and visits, this data is annotated to acknowledge that each record contains a varying degree of margin of error, which likely diminishes its capability to depict the real-world scenario and the relationship with trip demands. Second, this data is estimated based on census tracks, whose scale is larger than the block groups (the statistical unit for population density data we have collected). As a result, some errors may arise during the data reclustering process into the prediction regions. The above reasons likely deteriorate the model performance. However, there is potential for improvement if more precise, high-resolution socio-demographic data can be obtained.

Regarding management implications, E-scooter service providers can optimize fleet management and boost ridership using our real-time predictions of demand imbalances. Our proposed prediction model can highlight the regions with significantly higher demands in the next hour. By integrating with real-time E-scooter position tracking and central dashboard monitoring, the service operators can comprehensively understand the differences in supply and demand for each region and place idle E-scooters from the nearby regions with low demand to the regions with highly unsatisfied demands in advance. This short-term prediction is suitable for real-time E-scooter reallocation in coordination with nearby dispatch stations. Moreover, demand forecasts can play a pivotal role in shaping incentive strategies as suggested by others (for example, Yun et al., 2022), for specific regions and times that remain undersupplied despite reallocation efforts. These strategies may involve reducing rental fees to encourage users to return E-scooters to those regions in advance. Considering other practical scenarios, our proposed model can also be adjusted for longer-term predictions to accommodate the service operation schedulings (such as a 6-hour demand prediction for a 4-time-a-day shift schedule). Note that the hyperparameter of the number of past hourly data as inputs should be more than 6 h and be retrained to obtain. In this scenario, our research can assist the operators in developing the overall operation and maintenance strategies, including E-scooter reallocation, sanitation, inspection, and battery swapping. The operators can schedule different E-scooters alternately in high-usage regions to avoid excessive usage of any single one. Especially after the COVID-19 pandemic, E-scooter cleanliness (for instance, disinfecting the handles and cleaning the dirt) has been emphasized and has affected user willingness (Pourfalamatoun and Miller, 2023). Additionally, suitable maintenance strategies for inspection and battery swapping could extend E-scooters' lifespan (Severengiz et al., 2021). In this manner, unnecessary maintenance and labor costs can decrease while high-demand regions' service capacity and quality can increase.

Apart from the perspective of service operators, our research also contributes to establishing city policies related to E-scooter safety and parking issues for the city managers. The studies by Sikka et al. (2019) and Yang et al. (2020) indicate that the growing number of traffic crash records need attention. Yet, policies and regulations for this emerging micro-mobility mode still require enhancement. High-traffic regions can be highlighted based on temporal demand prediction, and the adjustment of traffic signals and the planning of sidewalks and bike lanes in those regions need to be reconsidered. Similarly, applying our proposed model to retrain and predict temporal drop-off demands can assist in identifying regions with high parking demands. Taking into account the overall development and land use in those regions, the government can replan appropriate parking spaces to reduce disorderly parking, sidewalk obstruction, and clutter issues (Zakhem and Smith-Colin, 2021), or even require operators to enhance the real-time removal of excess E-scooters.

6. Conclusion and future work

This study proposes SpDCGRU, a novel spatio-temporal graph neural network for real-time shared E-scooter trip demand prediction. To address the challenge of accurately predicting imbalanced and sparse trip demand, we merge diffusion convolution (DC) layers into the GRU model architecture to effectively capture spatio-temporal dependencies. Our experiments demonstrate that DC-based and GRU-based models outperform other convolution-based and LSTM-based models. Specifically, SpDCGRU achieves superior performance compared to GCRN_GRU; our model improves the MAE by 4.75%, the RMSE by 3.26%, the F1 score by

0.39%, and each AUC metric by 1.39%, 1.69%, 0.70%, 1.63%, 2.56%, and 1.23%. Notably, SpDCGRU demonstrates remarkable performance in predicting both the zero and high trip demand levels, which can assist service operators in focusing on the regions demanding over ten E-scooters in the next hour and reallocating fleets accordingly. We also further consider the average prediction performance over different periods. The results reveal that the average daily performance is nearly identical, while the average hourly performance experiences a performance decrease due to a dramatic increase in E-scooter daytime usage. It remains to be improved in future research.

To highlight the extremely sparse and imbalanced trip demand issue, we recommend using spatial data reclustering techniques to filter out the regions with zero trip demand and recluster neighboring spatial divisions. Furthermore, the results demonstrate that integrating the MAE loss and the focal loss with the loss penalty outperforms the conventional single MAE loss and cross-entropy loss functions. Notably, these strategies have enhanced the performance of different trip demand levels in different ways, with the loss penalty strategy mainly improving the zero trip demand by 12.80% and the focal loss mainly improving the high trip demand level by 2.69% compared to the method of the single MAE loss.

To improve the model's capability and capture urban complexity, we define comprehensive and detailed learning features based on built environments, weather conditions, and periodic features. Our results demonstrate that the periodic features play the most essential role in contributing to trip demand prediction, improving the MAE by 14.73%, the RMSE by 12.57%, and the F1 score by 2.70% in the ablation study. Weather conditions are the second predominant feature, improving the MAE by 6.65%, the RMSE by 9.24%, and the F1 score by 1.16%. Notably, periodic features have the greatest impact on the low trip demand level, while weather condition features play a more critical role in the high trip demand level. Features of population density and spatial relations are particularly important for AUC₃ and AUC₄ metrics, with improvements of 3.43% and 1.50%, respectively. POI and land-use entropy features are relatively less important but still contribute to the model's performance. Our practical experiments show that all learning features are crucial and should not be simplified.

While the proposed SpDCGRU model demonstrates exceptional performance in shared E-scooter trip demand prediction, there are still limitations and opportunities for improvement in future research. Our model mainly focuses on the trip demand prediction for one of the micro-mobility modes, the shared E-scooter. Forecasting trip demands for another similar transportation mode (such as shared bikes or shared moped scooters) could refer to our training framework and model as a reference; however, the prediction period should be defined according to the transportation mode and the application scenario. In this study, we discover the trip sparsity and imbalance and propose the SpDCGRU model accordingly. If our model is utilized for other cities, the reclustering and fusion-loss methods should be reconsidered, corresponding to the city's own trip characteristics. Besides, the model should also be retrained and fine-tuned to confirm the contribution of additional features. Regarding our future research, one major challenge that needs urgent attention is accurate prediction for the low trip demand level and accurate prediction during peak hours when encountering extreme imbalance trip data. Furthermore, incorporating detailed additional factors such as socio-demographic data (such as income level, employment status, sex, and age) (Xu et al., 2021), as well as weather conditions (such as air quality) on an hourly basis in each region, can assist in simulating trip behaviors more accurately.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.trd.2023.103962>.

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